



**MinersBuddy**

DGMS EXAM PREPARATION

COMPLETE EXAM SYLLABUS

# MINING MATE RESTRICTED

Metalliferous Mines | Opencast workings only

Common syllabus: Blaster (Restricted)

**13**

Modules

**145**

Topics

# Syllabus at a Glance

The Restricted syllabus covers mines having opencast workings only. The same syllabus applies to both Mining Mate and Blaster. Candidates moving to Unrestricted add the extra underground syllabus on top of this.

| NO.   | MODULE                                             | TOPICS |
|-------|----------------------------------------------------|--------|
| 1     | Mine Legislation                                   | 16     |
| 2     | Duties, Reports & Examinations of Mate and Blaster | 9      |
| 3     | Opencast Mining Practice & Safety                  | 23     |
| 4     | Explosives, Blasting & Shotfiring (Opencast)       | 55     |
| 5     | First Aid & Emergency Management                   | 15     |
| 6     | Miner's Diseases & Dust Hazards                    | 7      |
| 7     | Mine Plans & Sections                              | 9      |
| 8     | Mine Machinery (Surface and Opencast Basics)       | 4      |
| 9     | Traffic Rules in Mines                             | 1      |
| 10    | Mine Working & General Safety                      | 3      |
| 11    | Methods of Working: Opencast                       | 1      |
| 12    | Mine Lighting                                      | 1      |
| 13    | Abbreviations Used in Mining                       | 1      |
| Total |                                                    | 145    |

How to use: Part A is the base you must master first. Every topic below is examinable for the Restricted paper.

## 1 Mine Legislation

### Acts, Rules & Circulars

- The Mines Act, 1952
- The Mines Rules, 1955 (health, sanitation, leave)
- Mines Vocational Training Rules, 1966
- DGMS Circulars (technical and general), e.g. Tech. Circular 01/2024 on electronic detonators in opencast mines

### Metalliferous Mines Regulations, 1961 (core chapters)

- |                                                                           |                                           |
|---------------------------------------------------------------------------|-------------------------------------------|
| ■ Preliminary                                                             | ■ Means of access and egress              |
| ■ Returns, notices and records                                            | ■ Transport of men and materials: haulage |
| ■ Examinations and certificates of competency and fitness                 | ■ Mine workings                           |
| ■ Inspectors and mine officials                                           | ■ Explosives and shotfiring               |
| ■ Duties and responsibilities of workmen, competent persons and officials | ■ Machinery and plant                     |
| ■ Plans and sections                                                      | ■ Miscellaneous and Schedules             |

## 2 Duties, Reports & Examinations of Mate and Blaster

- |                                             |                                               |
|---------------------------------------------|-----------------------------------------------|
| ■ Application form for examination          | ■ Examinations / inspections by Mining Mate   |
| ■ Details of required experience            | ■ Precautions for flammable and harmful gases |
| ■ Subjects for Mate and Blaster examination | ■ Report of Mate                              |
| ■ Duties and responsibilities of a Blaster  | ■ Report of Blaster                           |
| ■ Duties and responsibilities of a Mate     |                                               |

## 3 Opencast Mining Practice & Safety

### Opencast fundamentals

- |                                                  |                                     |
|--------------------------------------------------|-------------------------------------|
| ■ Opencast (open) mines                          | ■ Arrangement of footpaths in mines |
| ■ Advantages and disadvantages of opencast mines | ■ Height and width of benches       |
| ■ Mine excavation / digging                      | ■ Spoil / debris dumps              |

### Statutory safety provisions

- |                                           |                                          |
|-------------------------------------------|------------------------------------------|
| ■ Laws for the safety of workers in mines | ■ Rules for movement of tubs / mine cars |
| ■ Laws for fencing and doors / gates      | ■ Dangers of surface railway wagons      |
| ■ Rules for haulage roadways              | ■ Fencing around opencast mines          |
| ■ Rules regarding travelling roadways     | ■ Roads for trucks and dumpers           |

### Operations and machines

- Supervision of work and maintenance of machines
- Precautions in drilling
- Precautions during blasting
- Precautions on opencast machines
- Safety functions of truck / dumper operators

### Types of fencing

- S-type fencing: S1, S2, S3
- G-type fencing: G1, G2, G3, G4, G5
- U-type fencing: U1, U2
- Types of opencast fencing and their use

## 4

## Explosives, Blasting & Shotfiring (Opencast)

### Explosives and detonators

- Explosives and their types
- Types of permitted explosives
- Explosives for opencast mines
- Liquid explosives (slurry / emulsion)
- Detonator (cap); ordinary detonator: advantages and disadvantages
- Electric detonator and delay detonator
- Detonating relay
- Detonating fuse (Cordtex)
- Types of fuse joints

### Statutory rules, storage and handling

- Appointment of Shotfirer and Blaster
- Conditions and precautions in using permitted explosives
- Blasting materials / equipment
- Magazine and precautions in magazine
- Rules regarding explosives in mines
- Safety instructions for shotfirer / blaster helpers
- Place of shelter
- Safety instructions for explosive carriers
- Transit slip

### Charging and firing

- General rules for drilling, charging, stemming and firing
- Exploder / blasting machine
- Stemming clay / material
- Methods of firing; firing with electric detonator
- Types of firing circuits and testing of circuit
- Firing of hole
- Zapro ohmmeter circuit tester

### Misfires

- Causes of and precautions against misfire
- Misfire in electric detonator
- Rules in the regulations for removing or extracting misfires
- Misfires in large diameter blasting
- General precautions

### Opencast blasting practice

- Blasting in opencast mines
- Work and precautions in opencast over underground mine
- Quantity of explosives
- Secondary blasting
- Methods of blasting; method of blasting in opencast
- Dangers and causes of explosive accidents in mines
- Deck charging (charging at different heights)
- Appendices 1 to 6
- Protection from flying rocks during blasting

### Advanced opencast blasting (reference: L. C. Kaku, Part Two)

- Principles, requirements and importance of blasting in opencast mines
- Practical aspects of explosive quality
- Methods of initiating explosives; sequential blasting machine
- Bulk transport of explosives
- Deep hole blasting and bulk mix explosive blasting
- Blast design: drilling pattern, firing pattern, delay interval, cord relay
- Special methods: sleeping hole, blasting in fire area, air deck, ANFO, cast blasting, pilot hole
- Controlled blasting: pre-splitting, smooth wall blasting, line drilling
- Control of fly rock, ground vibration and air blast; muffle blasting
- Nonel / shock tube system and electronic detonator
- Safety fuse blasting
- Blasting record and daily report of blasting overman
- Main causes of accidents from blasting in opencast mines
- Explosives Rules, 2008: an introduction
- Purchase of explosives with powder factor guarantee
- Duties of Magazine In-charge

## 5

### First Aid & Emergency Management

- Section A: Basic first aid techniques (CPR, recovery position, assessment)
- Section B: Respiratory system and respiration (choking, asthma, suffocation)
- Section C: Heart, blood circulation, shock (bleeding, wounds, amputation)
- Section D: Bones, joints and muscles (fractures, dislocations, sprains)
- Section E: Nervous system and unconsciousness (head injury, stroke, seizures)
- Section F: Gastrointestinal tract, diarrhoea, food poisoning and diabetes
- Section G: Skin, burns, fever, heat exhaustion, heatstroke and hypothermia
- Section H: Poisoning (ingestion, inhalation, injection)
- Section I: Bites and stings (animal, snake, insect)
- Section J: Senses (eyes, ears, nose, skin) and swallowed foreign objects
- Section K: Urinary and reproductive systems; emergency childbirth
- Section L: Psychological first aid
- Section M: Specific emergency situations and disaster management (triage)
- Section N: First aid techniques (dressings, bandages, stretchers, transport)
- Appendices: first aid kits and references

## 6

### Miner's Diseases & Dust Hazards

- Measurement of dustiness
- Dust control by the respiratory system
- Pneumoconiosis: classification of inorganic dusts, types of dust diseases
- Permissible dust concentrations
- Prevention and treatment of dust diseases
- Dust sampling methods and instruments
- Ailments of the ear (noise hazards)

## 7 Mine Plans & Sections

- Surface plan
- Geological plan
- Water danger plan
- Joint survey plan
- Electrical plan
- Combined / composite plan
- Manpower distribution plan
- Accident plan
- Conventions and symbols for preparing plans and sections

## 8 Mine Machinery (Surface and Opencast Basics)

- Wire ropes: composition, classification, flexibility and lay, constructions, selection and maintenance
- Belt conveyors
- Aerial ropeways: types, applicability, bi-cable ropeway components
- Mine pumps: centrifugal, turbine and other types

## 9 Traffic Rules in Mines

- Statutory traffic and transport guidelines in mines

## 10 Mine Working & General Safety

- Risk assessment
- Safety committees
- General mine safety

## 11 Methods of Working: Opencast

- Opencast extraction methods

## 12 Mine Lighting

- Mine illumination standards

## 13 Abbreviations Used in Mining

- Standard technical acronyms and definitions

## Reference Material

| REFERENCE                                                                                                        | SOURCE / AUTHOR                       |
|------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Mate and Blaster Goshti (Metal Mine Mining Examinations)                                                         | Arvind Kumar                          |
| Mine Explosives and Practical Blasting Techniques (Underground and Opencast)                                     | L. C. Kaku                            |
| Mine Machinery: A Comprehensive Guide                                                                            | Prof. Devidas S. Nimaje, NIT Rourkela |
| The Mines Act 1952, Mines Rules 1955, Metalliferous Mines Regulations 1961, Mines Vocational Training Rules 1966 | Statutory texts                       |
| First Aid and Emergency Management manual; Miner's Diseases and Dust Hazards; Mine Plans and Sections            | MinersBuddy content repository        |